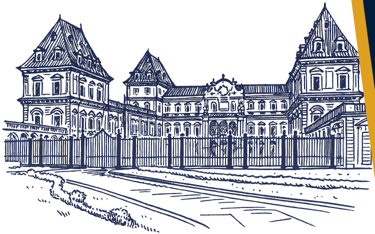
Politecnico di Torino

CrypTO
Crittografia e Teoria dei Numeri
DISMA - Politecnico di Torino



Random Self Reductions

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Cryptographic Proof Techniques 2025

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Random-self-reducible functions

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Problems

A *decision problem* is a set D of binary strings where we are asked if a given $x \in \{0, 1\}^*$ lies in D.
A *search problem* is a set S of binary strings (x, y) where given x we are asked to find an y such that $(x, y) \in \mathsf{S}$.

Reductions

A *poly-time reduction from problem* P_1 *to problem* P_2 ($\mathsf{P}_1 \leq_p \mathsf{P}_2$) is a poly-time algorithm transforming any instance x_1 of P_1 into an instance x_2 of P_2 such that $x_1 \in \mathsf{P}_1$ if and only if $x_2 \in \mathsf{P}_2$.

$$\begin{array}{ccc} P_1 & \xrightarrow{\quad A \quad} & P_2 \\ x_1 & & x_2 \end{array}$$

Nonadaptively k-rsr

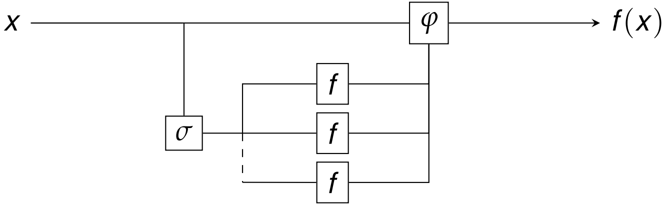
A function $f : \{0,1\}^* \to \{0,1\}^*$ is *nonadaptively k-rsr* if there are two poly-time functions σ, φ such that

1. for all n and all $x \in \{0,1\}^n$, for at least $3/4$ of all r in $\{0,1\}^{\omega(n)}$, we can compute $f(x)$ as
$$f(x) = \varphi\big(x, r, f(\sigma(1, x, r)), \ldots, f(\sigma(k, x, r))\big)$$
2. for all n, if r is an uniform random variable, then the random variables $\sigma(i, x_1, r)$ and $\sigma(i, x_2, r)$ are identically distributed for all $\{x_1, x_2\} \in \{0,1\}^n$ and all $i = 1, \ldots, k$.

$x \rightsquigarrow f(x)$ $\quad n = len(x)$

$k = k(n)$

r fair coin flips, $\omega(n)$

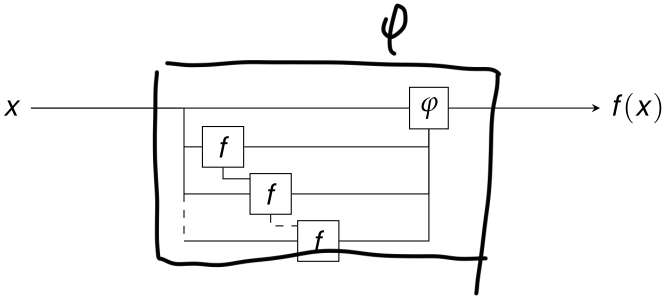


k(n)
randomised

Adaptively k-rsr

A function $f : \{0,1\}^* \to \{0,1\}^*$ is *adaptively k-rsr* if there is a poly-time oracle machine φ that, on input x of $\mathsf{len}(x) = n$, produces $k(n)$ rounds of oracle f-queries $y_i(x, r)$ to f, where the i-th query y_i might depend on previous queries and answers. The reduction φ must be such that

1. for all n and all $x \in \{0,1\}^n$, for at least $3/4$ of all r in $\{0,1\}^{\omega(n)}$, it outputs the correct answer $f(x)$.
2. for all n, if r is chosen uniformly at random, then the random variables $y_i(x_1, r)$ and $y_i(x_2, r)$ are identically distributed for all $\{x_1, x_2\} \in \{0,1\}^n$ and all $i = 1, \dots, k$, except if wrong answers were given on earlier rounds.

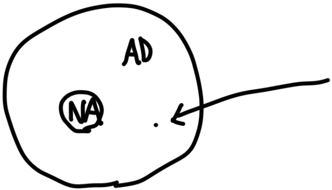

φ
x
φ
f(x)
f
f
f

Non-adaptive vs adaptive

$$k(n) = \mathrm{poly}(n)$$
$$w(n) = \mathrm{poly}(n)$$

Nonadaptively k-rsr:
Parallel queries, less powerful and smaller class.

Adaptively k-rsr:
Sequential queries, suitable for query dependence.



$$L \to L' \to L''$$
$$f = \chi_{L''}$$

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Discrete Logarithm problem

DLOG

- *Parameters:* a description $\Gamma = (\mathbb{G}, g)$ of the finite cyclic group group $\mathbb{G} = \langle g \rangle$ of order n.
- *Instance:* an element $h \in \mathbb{G}$.
- *Task:* find the unique $e \in \mathbb{Z}_n$ such that $g^e = h$.

Lemma

The $\mathsf{DLOG}_{\mathbb{G},g}$ *problem is poly-rsr.*

$x = h$

$f(x) = \mathrm{dlog}(x)$

σ?

φ?

$\boxed{\sigma}$ $\sigma(i, x, r) = h g^r$

$x = g^e$ $\quad y = \sigma(i, x, r)$

$\tilde{x} = g^{\tilde{e}}$ $\quad \hat{y} = \sigma(i, \tilde{x}, r)$

$\cdot\ z = g^a$

$$\mathbb{P}(\hat{y} = z) = \mathbb{P}(\tilde{x} g^r = z) = \mathbb{P}(g^{\tilde{e}+r} = g^a) = \mathbb{P}(r = a - \tilde{e}) = 1/|\mathbb{G}|$$

Φ $\varphi(x,r,f(\sigma(1,x,r))) = f(\sigma(1,x,r)) - r$

$= f(hg^r) - r$

$= d\log(hg^r) - r = (e+r) - r$

$K(n) - r \leq r$

$K = 1$

RSA Inversion problem

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RSAI

Consider an RSA modulus $N = pq$ with distinct primes p, q, and a public exponent e such that $\gcd(e, \varphi(N)) = 1$.

— *Parameters:* the system description (N, e).
— *Instance:* an element $x \in \mathbb{Z}_N^*$.
— *Task:* find the unique $y \in \mathbb{Z}_N^*$ such that $y^e \equiv x \mod N$

Lemma

The $\mathsf{RSAI}_{N,e}$ problem is poly-rsr.

instance x

$\boxed{\sigma}$ $\quad \sigma(i, x, r) = xr^e$

RSAI: $x = y^e \leadsto z = \sigma(i, x, r)$

RSAI: $\tilde{x} = \check{y}^e \leadsto \hat{z} = \sigma(i, \hat{x}, r)$

$\bar{x} = \bar{y}^e$

$$\mathbb{P}(\tilde{z} = \bar{x}) = \mathbb{P}(\tilde{x}r^e = \bar{y}^e)$$
$$= \mathbb{P}(\hat{y}^e r^e = \bar{y}^e)$$
$$= \mathbb{P}(r = \bar{y}/\tilde{y})$$
$$= 1/\varphi(N)$$

$\boxed{\varphi}$ $\varphi(x, r, f(\sigma(1,x,r))) = f(\sigma(1,x,r))/r$

$= f(x r^e)/r$

$= yr/r$ ✓

$f(x) = y$

$k(n) = 1$

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Elliptic curves

Definitions I

An *elliptic curve* is an algebraic variety

$$E = \left\{ (x, y) \in \bar{\mathbb{F}}_q^2 \cup \{\mathcal{O}\} \mid y^2 = x^3 + Ax + B \right\}$$

where the discriminant $\Delta = -16(4A^3 + 27B^2)$ is non-zero.
An *isogeny* is a surjective morphism $\psi : E \to E'$ such that for any $P, Q \in E$ we have $\psi(P + Q) = \psi(P) + \psi(Q)$.

$2P = \mathcal{O}$

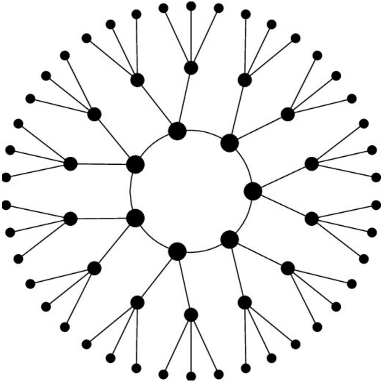


Figure: The 3-isogeny graph representing the isogeny class of the curves E with $j(E) = 607$ over F_{6007}. The different sizes of vertices represent different sizes of endomorphism rings. Michael Meyer, `https://doi.org/10.25972/OPUS-24682`, CC BY-SA 4.0.

Definitions II

The *endomorphisms* $\iota : E \to E$ of form a ring $\mathrm{End}(E)$. They include the "trivial" endomorphisms $P \mapsto k \cdot P$, which act as scalar multiplication by k.

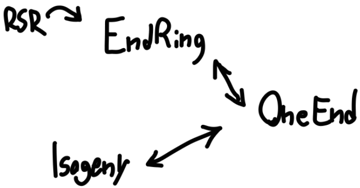
In our case, $\mathrm{End}(E)$ is a four-dimensional $\mathbb{Z}$-module.

$$\mathrm{End}(E) = \langle \underset{\substack{\uparrow \\ \text{trivial}}}{\varphi_1}, \varphi_2, \varphi_3, \varphi_4 \rangle$$

EndRing

Consider the finite field $\mathbb{F}_{q^2}$.

- *Parameters:* the field size q.
- *Instance:* a supersingular elliptic curve E over $\mathbb{F}_{q^2}$.
- *Task:* find the endomorphisms generating $\mathrm{End}(E)$ as a four-dimensional $\mathbb{Z}$-module.



Lemma

If we allow for the statistical indistinguishability of the distribution of the random variables $\sigma(i, x, r)$, EndRing$_q$ is poly-rsr.

EndRing: $E \overset{\sigma}{\rightsquigarrow} E_i'$?

$\sigma(i,x,r)$

1. ψ_i random isogeny (domain E)
2. $E_i' = \psi_i(E)$

$$
\begin{tikzcd}
E \arrow[r, "\psi_i"] \arrow[d] & E_i' \arrow[d] \\
E & E_i' \arrow[l, "\hat{\psi}_i"']
\end{tikzcd}
$$

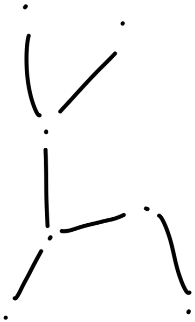
sampling ψ_i ?

- ψ_i represented via $\ker(\psi_i) \leq E$
- $\psi_i = \underbrace{\psi_i^{(2)} \circ \ldots\ldots \circ \psi_i^{(2)}}_{\ell}$

$\ell > \log_2 p^2 + \varepsilon$

$\sigma(i,x,r) = \psi_i(E) =: E_i'$

$f(E_i') = \langle l_1, l_2, l_3, l_4 \rangle$



2-isogenies

Group actions

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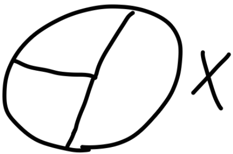
Group action

We say that $\mathbb{G}$ acts on X if there exists a map $\star : \mathbb{G} \times X \to X$, called *group action*, such that

— for every $x \in X$, we have $e \star x = x$, where e is the identity element in $\mathbb{G}$.

— for every $g, h \in \mathbb{G}$ and every $x \in X$, we have $(gh) \star x = g \star (h \star x)$.

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[d] & y \\ x' \arrow[r, "g'"'] & y' \end{tikzcd}$$



Transitivity

A group action is transitive if and only if it only induces one orbit on X. In other words, for any given $x, y \in X$ there exists some $g \in \mathbb{G}$ such that $y = g \star x$.

We require $X, \mathbb{G}$ to be finite and the existence of probabilistic poly-time algorithms performing the following operations:

1. **Membership.** Given a string g, decide if it represents an element of $\mathbb{G}$. Given a string x, decide if it represents an element of X.
2. **Equality.** Given $g, h \in \mathbb{G}$, decide if $g = h$.
3. **Random sampling.** Sample an element in $\mathbb{G}$ with given distribution $\mathcal{D}_G$ on $\mathbb{G}$.
4. **Group operations.** Given $g, h \in \mathbb{G}$, compute g^{-1} and gh.
5. **Action.** Given $g \in \mathbb{G}$ and $x \in X$, compute $g \star x$.
6. **Representation.** Given an element $g \in \mathbb{G}$ or $x \in X$, return the (not necessarily unique) representation as string in $\{0, 1\}^{\#\mathbb{G}}$.

GAIP

- *Parameters:* a group action $\star$ of $\mathbb{G}$ on X.
- *Instance:* two elements x, y of the set X.
- *Task:* find, if any, an element g such that $x = g \star y$.

$$y \xrightarrow[g]{} x$$

Lemma

If $\star$ is transitive, $\mathrm{GAIP}_{\mathbb{G},X,\star}$ is poly-rsr.

instance: (x,y)

$\boxed{\sigma}$ $\quad \sigma(i,(x,y),r) = (g_{i,x} \star x \,,\, g_{i,y} \star y) =: (x_i', y_i')$

$(x_i', y_i') = \sigma(i, (x,y), r)$
$(\tilde{x}_i', \tilde{y}_i') = \sigma(i, (\hat{x}, \tilde{y}), r)$

$z, w \in X$

$$\mathbb{P}((x_i', y_i') = (z,w)) = \mathbb{P}((g_{i,x} \star x, g_{i,y} \star y) = (z,w))$$
$$= \mathbb{P}(g_{i,x} \star x = z, g_{i,y} \star y = w)$$
$$= \mathbb{P}(\delta(z,x) = g_{i,x}, \delta(y,w) = g_{i,y})$$
$$= 1/|\mathbb{G}|^2$$

$$\varphi(x,y,r, f(\sigma(1,(x,y)))) = g_{1,x}^{-1} \star f(x_i', y_i') \star g_{1,y}$$

random instance

$$
\begin{tikzcd}
x \arrow[r, "g_{1,x}"] \arrow[d, "f(x,y)"'] & x'_i \arrow[d, "f(x',y')"] \\
y \arrow[r, "g_{1,y}"'] & y'_i
\end{tikzcd}
$$

$(i=1)$

$$g_{1,x}^{-1} * f(x'_1, y'_1) * g_{1,y}$$

LCE

Consider linear codes defined over $\mathbb{F}_q$.

- *Parameters:* code parameters n, k, q.
- *Instance:* the generator matrices G, G' of two (n, k)-codes C, C'.
- *Task:* find, if any, $S \in GL_k(\mathbb{F}_q)$ and $Q \in M_n(\mathbb{F}_q)$ such that $G' = SGQ$.

$Q = P \cdot D$

$= (\cdot \;\; \cdot \;\; \cdot)$

This is an instance of Group Action Inversion, where

$$\star : \mathbb{G} \times X \longrightarrow X$$
$$((S; (\alpha, Q)), M) \longmapsto S \cdot \alpha(MQ)$$

S inv.

α automorphism

Q monomial

$g \in \mathbb{G}$:

$g \star M = (S; (\alpha, Q)) \star M$

inv.

aut.

mon.

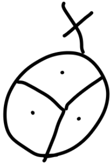
$$\mathbb{G} = GL_k(\mathbb{F}_q) \times (\mathrm{Aut}(\mathbb{F}_q) \ltimes M_n(\mathbb{F}_q))$$

Corollary

LCE is not poly-rsr.

The action is not transitive, two codes with different invariants cannot be mapped one into the other. Consider for instance

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \qquad G' = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$



over $\mathbb{F}_2$. They generate two $(4, 2)$-linear codes,

$$C = \{(1,0,1,0), (0,1,0,1), (1,1,1,1), (0,0,0,0)\}$$
$$C' = \{(1,0,1,1), (0,1,1,0), (1,1,0,1), (0,0,0,0)\}$$

with minimum distances are $d(C) = 2$ and $d(C') = 3$. The action of an element g such that $C' = g \star C = \phi(C)$ would be an isometry, meaning that $d(C') = d(\phi(C))$.

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One-way functions

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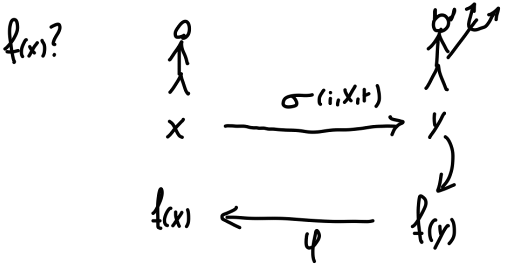
Angluin and Lichtenstein [2]: most classical security assumptions are based on random self-reducible problems, perhaps because the class of poly-rsr functions seems to yield one-way functions.

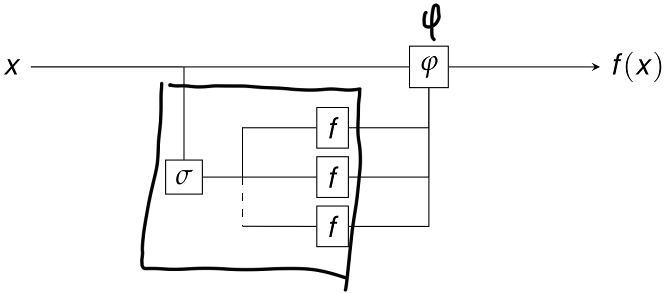
- Instance-hiding schemes
- Self-correction

not in AL

Application 1

Abadi et al. [1]: *instance hiding schemes*.





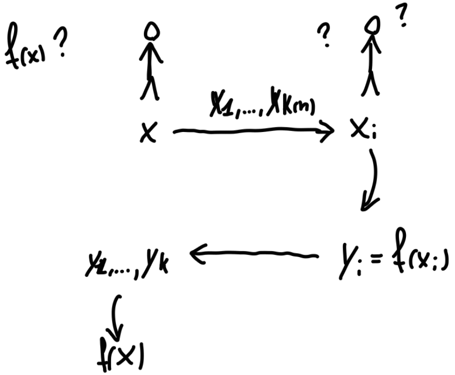
Corollary

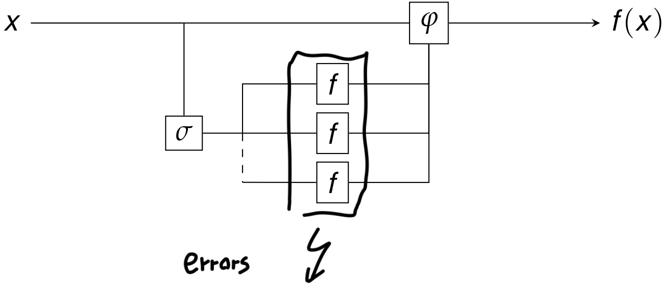
Since DLOG is nonadaptively 1*-rsr, then it is a* 1*-ihs (instance hiding scheme).*

k(n)

Application 2

Blum et al. [3]: *self-correction*.




x
φ
$f(x)$
f
σ
f
f
errors

Weak instances

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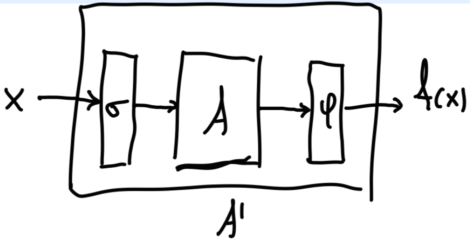
Corollary

Consider a poly-rsr f. If an algorithm $\mathcal{A}$ correctly solves $f(x)$ for a non-negligible fraction $\rho(n) > 0$ of uniformly random length-n inputs, we can construct a randomized algorithm $\mathcal{A}'$ that uses $\mathcal{A}$ as a subroutine to correctly evaluate f on all inputs with high probability.



Application 3

Protocol security.

- DLOG $\leftsquigarrow$?
- RSAI $\leftsquigarrow$?

Computational complexity

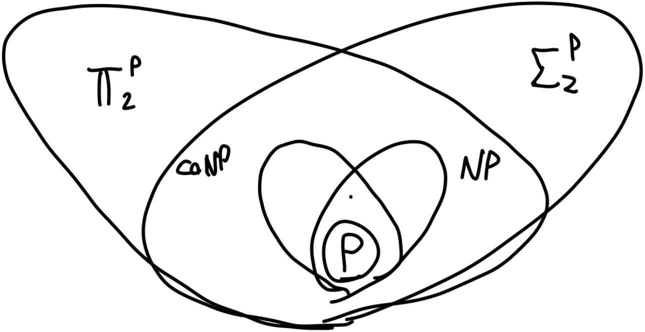
- P, all decision problems D that can be solved by a deterministic algorithm in poly-time.
- NP, all decision problems D that can be solved by a non-deterministic algorithm in poly-time.
- coNP, all decision problems D whose complement $\overline{\mathsf{D}} = \{x \mid x \notin \mathsf{D}\}$ is in NP.

For any complexity class C, a problem D is C-*hard* if every problem in C can be reduced to D in polynomial time, and if it also lies in C we call it C-*complete*.

$$\Sigma_0^P = \mathsf{P} \qquad \Pi_0^P = \mathsf{P} \qquad \Delta_0^P = \mathsf{P}$$

$$\Sigma_1^P = \mathsf{NP} \qquad \Pi_1^P = \mathsf{coNP} \qquad \Delta_1^P = \mathsf{P}^{\mathsf{NP}}$$

$$\Sigma_{i+1}^P = \mathsf{NP}^{\Sigma_i^P} \qquad \Pi_{i+1}^P = \mathsf{coNP}^{\Sigma_i^P} \qquad \Delta_{i+1}^P = \mathsf{P}^{\Sigma_i^P}$$



$$
\begin{aligned}
\Sigma_0^P &= \mathsf{P} & \Pi_0^P &= \mathsf{P} & \Delta_0^P &= \mathsf{P} \\
\Sigma_1^P &= \mathsf{NP} & \Pi_1^P &= \mathsf{coNP} & \Delta_1^P &= \mathsf{P}^{\mathsf{NP}} \\
\Sigma_{i+1}^P &= \mathsf{NP}^{\Sigma_i^P} & \Pi_{i+1}^P &= \mathsf{coNP}^{\Sigma_i^P} & \Delta_{i+1}^P &= \mathsf{P}^{\Sigma_i^P}
\end{aligned}
$$

$$
\begin{tikzcd}
 & \mathsf{coNP} = \Pi_1^{\mathrm{P}} \arrow[rr] \arrow[dr] & & \Pi_2^{\mathrm{P}} \arrow[rr] \arrow[dr] & & \Pi_3^{\mathrm{P}} \\
\Delta_0^{\mathrm{P}} = \Sigma_0^{\mathrm{P}} = \mathsf{P} = \Pi_0^{\mathrm{P}} = \Delta_1^{\mathrm{P}} \arrow[ur] \arrow[dr] & & \mathsf{P}^{\mathsf{NP}} = \Delta_2^{\mathrm{P}} \arrow[ur] \arrow[dr] & & \Delta_3^{\mathrm{P}} \arrow[ur] \arrow[dr] & \\
 & \mathsf{NP} = \Sigma_1^{\mathrm{P}} \arrow[rr] \arrow[ur] & & \Sigma_2^{\mathrm{P}} \arrow[rr] \arrow[ur] & & \Sigma_3^{\mathrm{P}}
\end{tikzcd}
$$

Application 4

Feigenbaum and Fortnow [4]: *collapse of the hierarchy*.

Theorem ([4])

If an NP*-complete function* f *is adaptively* $O(\log n)$*-rsr, then the polynomial hierarchy collapses at the third level.*

References

[1] Martin Abadi, Joan Feigenbaum, and Joe Kilian. "On hiding information from an oracle". In: *Proceedings of the nineteenth annual ACM symposium on Theory of computing*. 1987, pp. 195–203.

[2] Dana Angluin and David Lichtenstein. "Provable security of cryptosysterns: A survey". In: *Yale Univ., New Haven, CT, USA, Tech. Rep. TR-288* (1983).

[3] Manuel Blum, Michael Luby, and Ronitt Rubinfeld. "Self-testing/correcting with applications to numerical problems". In: *Proceedings of the twenty-second annual ACM symposium on Theory of computing*. 1990, pp. 73–83.

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[4] Joan Feigenbaum and Lance Fortnow. "Random-self-reducibility of complete sets". In: *SIAM Journal on Computing* 22.5 (1993), pp. 994–1005.

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